Discrete Math HW7

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1. Solve the system of linear equations using the Gaussian method.

$$\begin{cases} x + y - 2z = 6 \\ 2x + 3y - 7z = 16 \\ 5x + 2y + z = 16 \end{cases}$$

$$\begin{cases} x + y - 2z = 6 \\ 2x + 3y - 7z = 16 \\ 5x + 2y + z = 16 \end{cases} \rightarrow \begin{pmatrix} 1 & 1 & -2 \\ 2 & 3 & -7 \\ 5 & 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 6 \\ 16 \\ 16 \end{pmatrix} \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & -2 & 6 \\ 2 & 3 & -7 & 16 \\ 5 & 2 & 1 & 16 \end{array}\right] \rightarrow\rightarrow$$

$$\left[\begin{array}{ccc|c} 0 & -1 & 3 & -4 \\ 0 & 11 & -37 & 48 \\ 10 & 4 & 2 & 32 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 0 & 0 & -4 & 4 \\ 0 & 11 & -37 & 48 \\ 10 & 4 & 2 & 32 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 0 & 0 & 1 & -1 \\ 0 & 11 & -37 & 48 \\ 5 & 2 & 1 & 16 \end{array}\right] \rightarrow$$

$$\left[\begin{array}{ccc|c} 0 & 0 & 1 & -1 \\ 0 & 11 & 0 & 11 \\ 5 & 2 & 1 & 16 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 0 & 0 & 1 & -1 \\ 0 & 1 & 0 & 1 \\ 5 & 2 & 1 & 16 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 0 & 0 & 1 & -1 \\ 0 & 1 & 0 & 1 \\ 5 & 0 & 0 & 15 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 0 & 0 & 1 & -1 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 3 \end{array}\right]$$

Thus, $x = 3, y = 1, z = -1$

2. Find the eigenvectors and eigenvalues of a linear operator

$$A = \begin{pmatrix} 1 & 0 & 3 \\ 2 & 1 & 2 \\ 3 & 0 & 1 \end{pmatrix}$$

$det(A - \lambda I) = 0$

$|A-\lambda I| = \begin{vmatrix} 1-\lambda & 0 & 3 \\ 2 & 1-\lambda & 2 \\ 3 & 0 & 1-\lambda \end{vmatrix}$= (1-$\lambda$)^3 + 3(−3(1 − λ) = (1 − λ)^3 − 9(1 − λ) = (1 − λ)(λ^2 − 2λ − 8) = (1 − λ)(λ − 4)(λ + 2) = 0

Thus, eigenvalues are $\lambda_1 = 1, \lambda_2 = 4, \lambda_3 = -2$

$\lambda_1 = 1 \to (A - \lambda_1 I) \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$

$$\left(\begin{array}{ccc|c} 0 & 0 & 3 & 0 \\ 2 & 0 & 2 & 0 \\ 3 & 0 & 0 & 0 \end{array}\right) \to \left(\begin{array}{ccc|c} 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{array}\right) \to \left(\begin{array}{ccc|c} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{array}\right)$$

Thus, $x_1 = 0, x_2 \in \mathbb{R}, x_3 = 0$

If $x_2 = 1$, for $\lambda_1 = 1, v_1 = (0, 1, 0)$

$\lambda_2 = 4 \to (A - \lambda_2 I) \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$

$$\left(\begin{array}{ccc|c} -3 & 0 & 3 & 0 \\ 2 & -3 & 2 & 0 \\ 3 & 0 & -3 & 0 \end{array}\right) \to \left(\begin{array}{ccc|c} -1 & 0 & 1 & 0 \\ 2 & -3 & 2 & 0 \\ 1 & 0 & -1 & 0 \end{array}\right) \to \left(\begin{array}{ccc|c} -1 & 0 & 1 & 0 \\ 2 & -3 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right) \to$$

$$\left(\begin{array}{ccc|c} -2 & 0 & 2 & 0 \\ 0 & -3 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right) \to \left(\begin{array}{ccc|c} -1 & 0 & 1 & 0 \\ 0 & -3 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right) \to \begin{cases} -x_1 + x_3 = 0 \\ -3x_2 + 4x_3 = 0 \end{cases} \to \begin{cases} x_1 = x_3 \\ x_2 = \frac{4}{3}x_3 \\ x_3 \in \mathbb{R} \end{cases}$$

If $x_3 = 1$, then for $\lambda_2 = 4$, eigenvector $v_2 = (1, \frac{4}{3}, 1)$

$\lambda_3 = -2 \to (A - \lambda_3 I) \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$

$$\left(\begin{array}{ccc|c} 3 & 0 & 3 & 0 \\ 2 & 3 & 2 & 0 \\ 3 & 0 & 3 & 0 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right) \to \begin{cases} x_1 + x_3 = 0 \\ x_2 = 0 \end{cases} \to \begin{cases} x_1 = -x_3 \\ x_2 = 0 \\ x_3 \in \mathbb{R} \end{cases}$$

If $x_3 = 1$, then for $\lambda_3 = -2$, eigenvector $v_3 = (-1, 0, 1)$

3. How many integers between 0 and 999 contain

   a) exactly one digit 7?

Since there are 3 digits, and only one of them is 7, total number of integers with exactly one seven is $3 * 9^2 = 243$

b) at least one digit 7?

Since there are 1000 integers between 0 and 999 and 9^3 integers that does not contain 7, the number of integers that contain and least one 7 is $1000 - 9^3 = 271$

4. How many numbers less than one million can be written using the digits 8 and 9?

   Since the largest possible number is 999999, all integers must contain at most 6 digits.

   For 1 digit numbers there are 2^1 possible numbers - 8,9

   For 2 digit numbers there are 2^2 possible numbers - 88,99,89,98

   ...

   For 6 digit numbers there are 2^6 possible numbers

   So the total number of such integers is $2 + 2^2 + 2^3 + 2^4 + 2^5 + 2^6 = 126$

5. How many four-digit numbers greater than 2100 can be formed from the digits 1, 2, 3, 4, 5, 0 if no digit is repeated in the number?

   N of such integers = n (numbers starting with 2 and greater than 2100) + n (numbers starting with 3,4, or 5, and greater than 2100) = $(1 * 4 * 4 * 3) + (3 * 5 * 4 * 3) = 48 + 180 = 228$